



Dynamic Response of Potato Growth Indicators to Effective Cumulative Temperature and Water and Nitrogen Regulation Based on a Logistic Model

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Abstract

Potato is one of the key food crops and China is the largest potato producer in the world. Taking into account the combined effects of irrigation and nitrogen fertilization with effective accumulated temperature, it becomes essential to optimize the levels of irrigation and nitrogen application for agricultural production. The objective of this study is to examine the prediction model for the growth dynamics of potato leaf area index (LAI) and biomass. We aim to analyze the characteristic parameters of these variables under various water and nitrogen regulations. The analysis will be based on effective accumulated temperature. In the field experiment, different irrigation quota and nitrogen application rates were set, designated as W and N respectively. The LAI, biomass, and yield quality indicators of potatoes were measured, and the dynamic equations of potato LAI and biomass based on effective accumulated temperature under different conditions of water and nitrogen regulation were fitted using a logistic model. The results indicated that the LAI and biomass of potato followed a unidirectional incremental pattern in response to effective accumulated temperature, represented by an S-shaped curve. The dynamic change process showed a slow-fast-slow trend. In the W3N2 (2100 m³ ha⁻¹, 190 kg ha⁻¹) treatment, the highest LAI value for potato was observed at 1700 °C d. Excessive water and nitrogen could increase the demand for effective standing temperature in potato biomass. The treatment with 2100 m³ ha⁻¹ water and 110 kg ha⁻¹ nitrogen resulted in the highest average total yield of 54.09 t ha⁻¹ and commercial potato yield of 48.16 t ha⁻¹. The logistic model was employed for both calibration and validation purposes. The correlation coefficients (R^2) between the measured and simulated values, under moderate water and nitrogen conditions, exceeded 0.95. The study concluded that the logistic model effectively simulated the dynamic growth of potato under the regulation of effective accumulated temperature, water, and nitrogen. This model provides a theoretical basis for quantitatively simulating the growth and development dynamics of potato in the dry zone of Ningxia.

Keywords Effective accumulated temperature · Irrigation quota · Logistic model · Nitrogen application · Potato growth

Introduction

Potato, being a significant food crop (Ayadi et al. 2022; Fiers et al. 2012), plays a crucial role in ensuring food security (Scott et al. 2019). The potato crop's tuber yield and biomass improvements are strongly influenced by irrigation and fertilization (Ierna and Mauromicale 2018). Global crop production faces a significant constraint due to water scarcity, impacting potato growth in arid and semi-arid regions (Rady et al. 2021); (Xing et al. 2022). Nitrogen fertilizer plays a vital role in stimulating plant growth and enhancing crop yield (Li et al. 2018) and is the major yield-limiting nutrient in most agroecosystems (Ma et al. 2021). To achieve sustainable agricultural development, it is crucial to consider the synergistic effects of irrigation and nitrogen fertilization and optimize their application (Meise et al. 2019; Yang et al. 2023). Splitting nitrogen (N) fertilizer application is a key method to increase potato tuber yield and nitrogen fertilizer uptake efficiency (Silva et al. 2023). The scientific assessment of the amount and timing of grower inputs by technology for irrigation and fertilizer nitrogen management has become an important challenge in the process (Lo et al. 2019). The leaf area index and biomass are important indicators for simulating crop growth (Tang et al. 2022), and they show regular dynamic changes with the fertility of crops. However, the practical application of crop simulation using fertility as a step unit varies according to regional differences and the length of fertility (Shabani et al. 2014). The accumulated temperature is the most important meteorological factor affecting crop development, agricultural decision-making, yield estimation (Chu et al. 2016), and simulating crop growth. Although it is theoretically stable, the effective accumulated temperature required for different stages of fertility, years, regions, and varieties of crops is significantly different (Li et al. 2020). The effective temperature required at different stages of plant growth increases, and the mulch can decrease the proportion of effective temperature distribution in the growing period (Li et al. 2023a, b). Therefore, it is crucial to use effective accumulated temperature as a measure for simulating potato growth and development, rather than relying solely on fertility days. Taking into account the combined effects of irrigation and nitrogen fertilization with effective accumulated temperature, it becomes essential to optimize the levels of irrigation and nitrogen application for agricultural production.

The logistic model can reveal the universal law of developmental changes of an object under a limited environment and resources, and it can describe the complex dynamic developmental characteristics of plant growth because a logistic model has a simple structure (Yin et al. 2018). This concept finds broad application in diverse fields, such as agriculture, economy, medicine, scientific research, and more. Its usefulness lies in describing and forecasting the stages of growth and development, covering the occurrence, progression, and maturity of various phenomena. While the rate of change in the occurrence stage is low, the rate of change in the development stage is high, and the rate of change in the maturity stage tends to be low (Chen et al.

2009). Sepaskhah et al. (2011) used water and nitrogen as key factors affecting crop growth in the logistic model; they grew maize under different water and nitrogen management conditions and achieved better results. Liao et al. (2022) considered the combined effects of temperature and light on maize growth and constructed a modified logistic growth equation that was more effective in simulating maize growth. Zhang et al. (2022a, b) found that maize biomass allocation patterns are particularly important, and they used logistic models to study crop yield, plant growth rate, and plant biomass trends in different environments and continuous growth to provide new insights into yield prediction in the early growth stages. Giacomini Sari et al. (2019) found that logistic growth models can be used instead of ANOVA for analyzing the empirical data obtained from horticultural crops. Logistic models predicted branch growth in wheat better than Gompertz models under different tillage and residue management practices (Mohanty et al. 2006). Additionally, the model can be applied to assess ecological technology (Hu et al. 2022). Jahangirlou et al. (2023) used a three-parameter logistic model to calculate the starch and oil content of grains and showed that the model could predict the protein, starch, and oil content in maize kernels.

Most studies on logistic models investigated maize, tomato, wheat, and other crops, but studies on the simulation of potato growth and development, based on logistic models with different water and nitrogen regulation and effective temperature accumulation, are limited. As water and nitrogen strongly affect potato growth, it is necessary to determine the simulation model of the growth and development of potatoes under the control of water and nitrogen, and it is important to improve the prediction of crop growth and guide the control of crop production. Consequently, by examining various combinations of water and nitrogen treatments and their impact on the leaf area index (LAI) and biomass of potato under effective accumulated temperature conditions, we formulated a dynamic model. This model specifically addresses the relationship between effective accumulated temperature and the dynamic behavior of potato LAI and biomass during the crucial fertility period under diverse water and nitrogen conditions. The model accurately predicted the LAI and biomass of potatoes. The study results provide data support and a theoretical basis for the quantitative analysis of potato growth dynamics using effective cumulative temperature.

Materials and Methods

Location of the Study

The field trials were conducted from 2021 to 2022 at the designated trial site located in Yu Wang Town, Concentric County, Ningxia Hui Autonomous Region (36°48'2" N, 106°21'53" E, altitude 1489 m), as shown in Fig. 1a. The soil in the experimental area was classified as sandy loam, and the soil characteristics within the 0–60-cm depth are provided in Table 1. The field meteorological data were obtained from a compact automatic weather station installed within the experimental field, as depicted in Fig. 2.

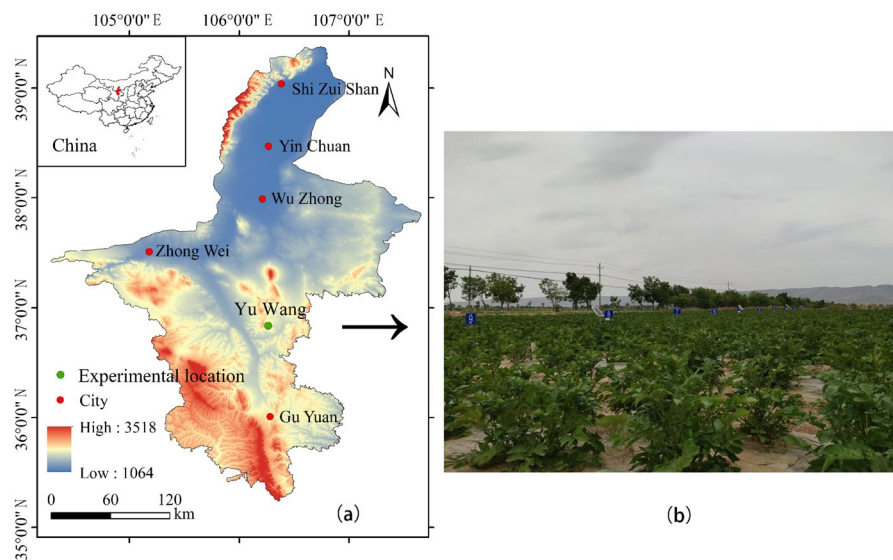


Fig. 1 Location of the study site and planting map

Experimental Design

This experiment was conducted based on the local actual situation and with reference to the Ningxia Potato Drip Irrigation Planting Technical Regulations issued by the Department of Water Resources of Ningxia Hui Autonomous Region in 2017. Treatment settings for irrigation were as follows: conventional irrigation ratio ($2100 \text{ m}^3 \text{ ha}^{-1}$, W3), 20% water savings from conventional irrigation ($1680 \text{ m}^3 \text{ ha}^{-1}$, W2), and 40% water savings from conventional irrigation ($1260 \text{ m}^3 \text{ ha}^{-1}$, W1). Treatment settings for nitrogen application were as follows: conventional nitrogen application (270 kg ha^{-1} , N3), 30% less than conventional nitrogen application (190 kg ha^{-1} , N2), and 60% less than conventional nitrogen application (110 kg ha^{-1} , N1). Each treatment was replicated three times, resulting in a total of 27 plots. Urea (46% N) was used as the nitrogen fertilizer and was applied using integrated water fertilizer technology. The division of fertility, irrigation volume, and the percentage of nitrogen application are presented in Fig. 3.

The potatoes were sown on April 30, 2021, and May 1, 2022, and harvested on October 2, 2021, and October 3, 2022. The variety used in the experiment was ‘Qingshu9’ which was planted using drip irrigation (Fig. 1b). The planting arrangement involved ‘one film and two rows’ with equal spacing, utilizing a plastic film that was 140 cm wide, and the seeds were buried 10 cm deep. The potato planting spacing was 50 cm, with a row spacing of 60 cm. Each row had 10 plants, and each plot contained 40 potatoes, resulting in a potato planting density of $33,345 \text{ plants ha}^{-1}$. The experimental plots were all 17.6 m^2 in area, with dimensions of 5.5 m in length and 3.2 m in width. These plots were surrounded by protection rows, with inter-area protection rows having a width of 1 m and peripheral protection rows

Table 1 Soil characteristics and basic nutrient parameters

Soil layers cm	Bulk density g cm ⁻³	Wilting point values g g ⁻¹	Field water holding capacity g g ⁻¹	pH	total salt g kg ⁻¹	Organic matter g kg ⁻¹	Total Nitrogen g kg ⁻¹	Total Phospho- rus g kg ⁻¹	Total Potassium g kg ⁻¹	Fast-acting phosphorus mg kg ⁻¹	Fast-acting potassium mg kg ⁻¹
0–20	1.41	0.10	0.19	7.70	0.37	0.44	0.52	15.51	7.78	11.90	106.00
20–40	1.38	0.10	0.20	7.86	0.30	0.44	0.53	15.80	7.49	12.40	106.00
40–60	1.36	0.12	0.23	8.03	0.27	0.36	0.49	14.45	6.08	7.20	68.00

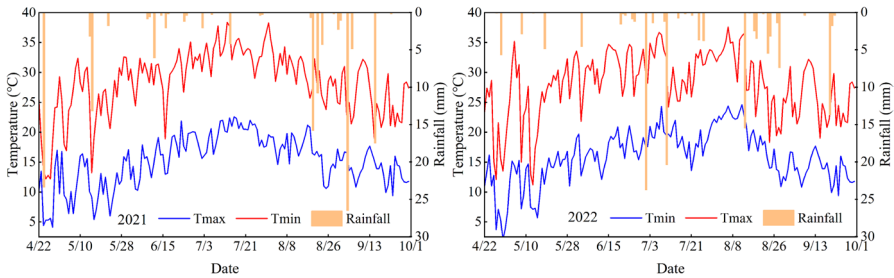


Fig. 2 The temperature and rainfall recorded during the potato fertility period in the study area

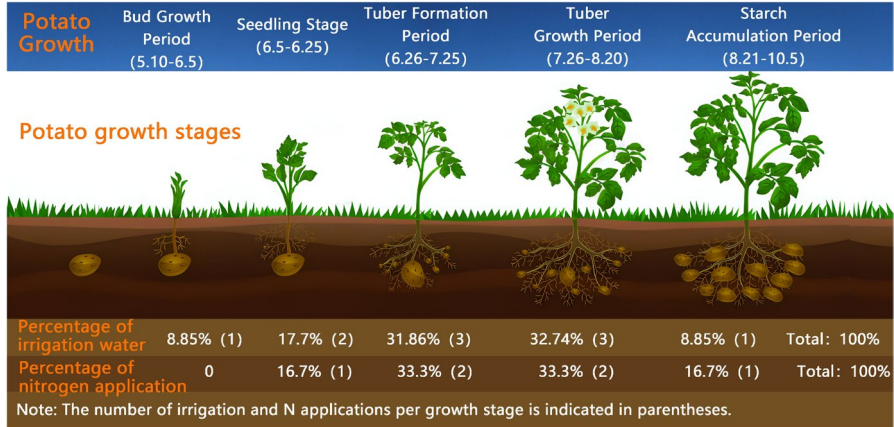


Fig. 3 Allocation ratio of irrigation and N application for each fertility period of potato

having a width of 2 m. A total of 27 plots were set up, and each treatment had three replications in the experiment. Protective rows were established around each plot. Prior to sowing, high-quality potato seeds were selected and mixed with rock flour. The sources of irrigation water were cisterns and wells. Drip irrigation for the potatoes was carried out using a patch-type drip irrigation belt with a 16-mm diameter, operating at a pressure of 0.1 MPa, with a low head spacing of 50 mm, and a flow rate of 2.2 L/h. Each plot was equipped with an individual water meter. Agronomic measures such as weeding were implemented following local routines.

Measurement Items and Methods

Biomass and the Leaf Area Index

During each reproductive period of the potato plant, samples were collected from each plot, with three plants per plot and three replications. The potato stems, leaves, roots, and tubers were carefully separated and individually dried in a blast drying oven. After reaching an effective accumulated temperature, the samples were

weighed to calculate the biomass of the potatoes (Qi et al. 2020). Simultaneously, potato leaves were subjected to weighing through punching to determine the leaf area of the potatoes (Jonckheere et al. 2004). This measurement was used to calculate the LAI (Wang et al. 2019).

$$\text{LAI} = \frac{\text{the total area of the leaf}}{\text{the occupied land area}} \quad (1)$$

Yield

Once the potatoes reached maturity, a total of 10 potato plants were excavated from each plot and weighed to determine the yield per plant. Subsequently, the total yield per hectare for different treatments was computed based on the planting density. Commercial potatoes were classified as tubers with a mass exceeding 75 g.

Quality

The vitamin C content and reducing sugar content of the potatoes were assessed using the titration method, while the starch content was determined using the acid hydrolysis method (Xing et al. 2020).

Effective Cumulative Temperature

The effective cumulative temperature T_i for this experiment was calculated as follows:

$$T_i = \sum (T_{i\text{avg}} - T_{\text{base}}) \quad (2)$$

Here, T_i indicates the cumulative effective temperature ($^{\circ}\text{C d}$) from day 1 to day i after sowing; $T_{i\text{avg}}$ indicates the average daily temperature ($^{\circ}\text{C}$); T_{base} indicates the lower biological limit temperature of the crop ($^{\circ}\text{C}$). The biological starting temperature of potatoes is 10°C (Meno et al. 2020; Yuchuan et al. 2017).

Logistic Model Characteristic Parameters and Validity Test

General Equation

The growth pattern of the potatoes transitioned from an initial slow phase to a rapid phase, and then reverted back to a slower phase, following a characteristic ‘S’ curve. This trajectory corresponds well with the predictions of the logistic growth model.

$$Y = \frac{K}{1 + ae^{-bx}} \quad (3)$$

where Y represents the LAI and total biomass of potatoes, while K represents the maximum growth limit of LAI and biomass for individual potato plants under specific environmental conditions. The parameter ‘ a ’ corresponds to the base state

parameter; e is the natural logarithmic base, b denotes the growth rate coefficient, and x denotes the effective cumulative temperature.

Growth Rate Equation

Derivation of the logistic equation, i.e., Eq. (3), yields the growth rate equation:

$$V = \frac{Kabe^{-bx}}{(1 + ae^{-bx})^2} \quad (4)$$

Determination of the Eigenvalues of the Equations

The maximum growth rate V_1 and the effective cumulative temperature T_1 required to reach the maximum growth rate can be determined by taking the first-order derivative of Eq. (4) and making it equal to 0. The equations are as follows:

$$V_1 = \frac{Kb}{4} \quad (5)$$

$$T_1 = \frac{\ln a}{b} \quad (6)$$

The two inflection points on the growth curve can be determined by taking the second-order derivative of Eq. (4) and making it equal to 0. The equations are as follows:

$$T_2 = \frac{\ln a - \ln(2 + \sqrt{3})}{b} \quad (7)$$

$$T_3 = \frac{\ln a - \ln(2 - \sqrt{3})}{b} \quad (8)$$

The average growth rate of the fast growth period can also be determined from T_2 and T_3 using Eq. (9):

$$V_2 = \frac{K}{\sqrt{3}(T_3 - T_2)} \quad (9)$$

Here, the three crucial temperature thresholds in the potato growth curve are denoted as T_1 , T_2 , and T_3 . These thresholds represent the maximum growth rate, the cumulative temperature required to initiate the fast-growth phase, and the temperature threshold for the slow-growth phase, respectively. Specifically, the growth stages can be categorized as follows: 0– T_2 for the period of incremental growth, T_2 – T_3 for the period of rapid growth, and T_3 –maturity for the period of slow growth.

Model Validity Test

To verify the fit between the simulated results of the logistic model and the measured results, we evaluated the root mean square error (RMSE), the index of fit (d), the normalized root mean square error (NRMSE), and the coefficient of determination (R^2) to test the validity of the model using Eqs. (10)–(13) (Eeswaran et al. 2021).

$$\text{RMSE} = \sqrt{\sum_{i=1}^n \frac{(S_i - M_i)^2}{n}} \quad (10)$$

$$\text{NRMSE} = \sqrt{\sum_{i=1}^n \frac{(S_i - M_i)^2}{n}} \times 100\% / \bar{M} \quad (11)$$

$$d = 1 - \frac{\sum_{i=1}^n (S_i - M_i)^2}{\sum_{i=1}^n \left(\left| S_i - \bar{M} \right| + \left| M_i - \bar{M} \right| \right)^2} \quad (12)$$

$$R^2 = \frac{\left[\sum_{i=1}^n (S_i - \bar{S})(M_i - \bar{M}) \right]^2}{\sum_{i=1}^n (S_i - \bar{S})^2 \sum_{i=1}^n (M_i - \bar{M})^2} \quad (13)$$

Here, S_i indicates the simulated value, M_i indicates the measured value, n indicates the number of measurement samples, and $\bar{M}$ indicates the measured average value.

Smaller values of RMSE and NRMSE indicate that the simulation results are more accurate. When the value of d tends to 1, the simulation results fit the measured results better. When the value of R^2 is close to 1, it indicates that the model is credible, and when the value of R^2 is close to 0, it is indicative of no significant relationship.

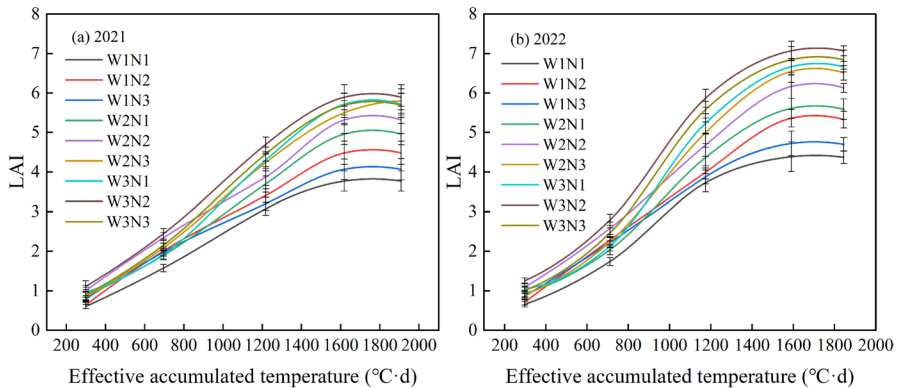
Data Analysis

We used Excel for data collation and averaging and the SPSS 26 software for model fitting and conducting ANOVA. A logistic nonlinear growth model was selected to fit the model parameters a , b , and k optimal estimates, where the effective cumulative temperature and growth indices were calculated using relative values, and Origin 2022 was used to plot the growth curves of the measured and model fitted values.

Table 2 Correlation analysis of potato LAI and total biomass under different water and nitrogen regulation

Year	Treatment								
	W1N1	W1N2	W1N3	W2N1	W2N2	W2N3	W3N1	W3N2	W3N3
2021	0.9882**	0.9826**	0.9913**	0.9937**	0.9877**	0.9966**	0.9959**	0.9902**	0.9920**
2022	0.9881**	0.9829**	0.9921**	0.9859**	0.9849**	0.9947**	0.9956**	0.9891**	0.9958**

The number of samples is 10. ** indicates significance at 0.01 level

**Fig. 4** Dynamic changes of LAI with different treatments

Results

Correlation Analysis of Potato LAI and Total Biomass

An analysis correlating the potato LAI with the total biomass for various treatments was conducted, and the results are delineated in Table 2. The results showed that potato LAI and total biomass under different water and nitrogen control showed consistent growth from the early stage to the stabilization period, and the correlation between them was above 0.95, which suggested a highly significant positive correlation, indicating that potato LAI and total biomass can be fitted to the same model in model simulation studies.

Characteristics of Potato LAI Changes

Dynamics of Potato LAI with Effective Accumulated Temperature and Its Modeling

The changes in potato LAI with effective accumulated temperature for each treatment are depicted in Fig. 4. The change trends of potato LAI under different water and nitrogen regulations were basically the same. All treatments showed a dynamic

trend of slow growth, then rapid growth, and finally transitioned into slow growth. The potato LAI was similar in 2021 and 2022 under various treatments. The highest LAI value of 6.48 was observed at 1700 °C d, corresponding to the W3N2 treatment. The W1 irrigation regime showed a steady increase in LAI in response to nitrogen application, with a constant effective cumulative temperature of 300 °C d. In contrast, the W2 and W3 irrigation conditions exhibited an initial rise followed by a decrease as nitrogen application increased. Moreover, each treatment showed an LAI increase when the irrigation volume was augmented under equal nitrogen application parameters. Excessive nitrogen application was found to reduce the leaf area index, adversely affecting potato growth, while increasing irrigation water to adequate levels promoted LAI growth.

A non-linear regression analysis of the logistic model, correlating potato LAI with the cumulative temperature for each treatment in 2021, was conducted and is presented in Table 3. It was observed that the logistic model, based on effective cumulative temperature, could accurately replicate the dynamic growth trajectory of potato LAI under each distinct regime of water and nitrogen treatment.

Testing the Potato LAI Growth Model

The precision of the model simulations was tested using the potato leaf area index evaluated in 2022 and the cumulated temperature data. The determination coefficient (R^2) between the measured and simulated values among treatments ranged from 0.9886 to 0.9986 (Table 4); i.e., all coefficients were greater than 0.9886. The observations suggest a high level of reliability for the model, with the RMSE ranging between 0.019 and 0.261, and the NRMSE varying between 2.9133 and 5.9514%. Given that both values are below 10%, these findings indicate a proficient simulation quality. The goodness-of-fit index (d) was above 0.99, indicating that the simulation results fit very well with the measured results. Thus, the values and changes were similar under each treatment condition for the 2 years of the study, and the regression equation obtained for any year could be used in another year.

Table 3 Equation parameters of potato LAI for different water nitrogen treatments (2021)

Treatment	Parameters			R^2	RMSE	NRMSE (%)	d
	K	a	b				
W1N1	1.0444	14.4660	6.2368	0.998	0.0539	2.1078	0.9995
W1N2	1.0885	11.1283	5.1050	0.980	0.5050	17.0599	0.9666
W1N3	1.0521	9.7334	5.5420	0.993	0.1032	3.6630	0.9983
W2N1	1.0908	12.4980	5.2976	0.995	0.1223	3.7326	0.9986
W2N2	1.1037	9.3966	4.7944	0.989	0.1702	4.7679	0.9974
W2N3	1.0896	15.2060	5.5135	0.994	0.1464	3.9020	0.9986
W3N1	1.0870	15.5528	5.6060	0.994	0.1537	4.1300	0.9985
W3N2	1.0627	12.1069	5.6994	0.997	0.1064	2.6575	0.9992
W3N3	1.0632	15.1033	5.9184	0.997	0.1058	2.8049	0.9992

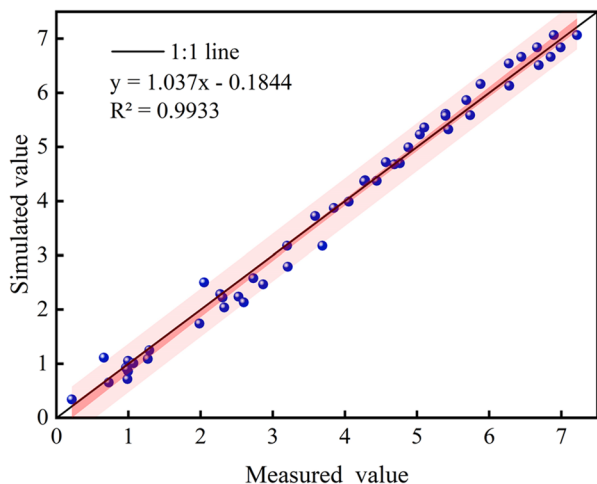
Table 4 Test of validity of measured and simulated values of potato LAI in 2022

Treatment	RMSE	NRMSE (%)	<i>d</i>	<i>R</i> ²
W1N1	0.1401	4.6624	0.9977	0.9941
W1N2	0.1757	4.9250	0.9975	0.9940
W1N3	0.1011	3.0689	0.9988	0.9978
W2N1	0.1806	4.8302	0.9977	0.9926
W2N2	0.1770	4.2454	0.9979	0.9949
W2N3	0.2027	4.7478	0.9980	0.9938
W3N1	0.0190	2.9133	0.9993	0.9986
W3N2	0.2610	5.9514	0.9968	0.9886
W3N3	0.2272	4.6768	0.9976	0.9925

The measured and simulated values of each treatment equation were plotted 1:1, as shown in Fig. 5. As depicted in the figure, the R^2 for LAI was 0.9933, signifying a high degree of simulation accuracy. This observation suggests that the logistic equation derived from the simulation holds biological relevance within the experimental conditions. It further indicates that dynamic alterations in potato LAI in response to cumulative temperature under various water and nitrogen regulations might be subject to more in-depth analysis.

Analysis of the Potato LAI Growth Rate and Its Equation Characteristic Parameters

The first-order derivatives of the logistic model obtained for each treatment in 2021 were used to obtain the equation for the LAI growth rate, which is graphically depicted in Fig. 6. At the W1 level, the potatoes were in the tuber formation stage when the effective cumulative temperature reached 694.6 °C d, during which $N1 > N3 > N2$, which indicated that potatoes did not need large quantities of

Fig. 5 The 1:1 plot of measured and simulated potato LAI in 2022

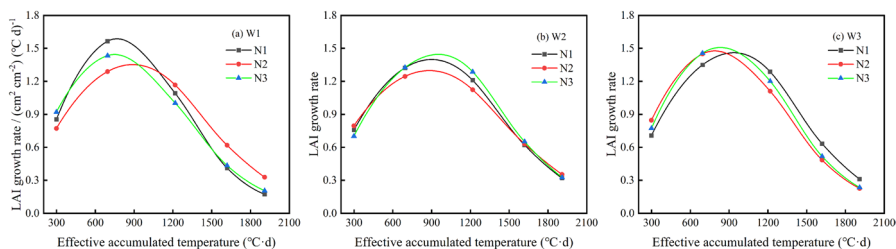


Fig. 6 Changes in potato LAI growth rate under different water N treatments (2021)

nitrogen fertilizer during the tuber formation period when irrigation was low. After the effective cumulative temperature reached 1200 °C d, potatoes treated with moderate amounts of nitrogen (N2) showed a higher LAI growth rate. At the W2 level, the LAI growth rate under the N3 treatment was significantly greater than the other out, indicating that appropriate irrigation conditions can improve the efficiency of nitrogen fertilizer and promote the LAI growth rate. At the W3 level, the potato was in the tuber formation stage when the effective cumulative temperature reached 694.6 °C d, during which $N2 > N3 > N1$. This indicated that the potato plants needed proper N under intensive irrigation during tuber formation, and after the effective cumulative temperature reached 1200 °C d, the LAI showed the highest value in the N1 treatment, indicating that irrigation influenced the LAI more effectively than the applied N.

Table 5 presents the outcomes of the characteristic parameters derived from the logistic model, utilizing data from the potato LAI in 2021. Under the W1 treatment, as nitrogen application increased for each regime, the pattern $N1 > N3 > N2$ was observed. The T1 time point required to achieve the maximum growth rate was most significant for the N2 treatment, clocking in at 900.96 °C d. The T2 cumulative temperature marked the potato tuber formation phase, and the effective cumulative

Table 5 The characteristic parameters of the logistic model for the changes in potato LAI (2021)

Treatment	Rate peak parameter		Rapid increase period parameters		
	V1 ($\text{cm}^2 \text{cm}^{-2}$) (°C d) ⁻¹	T1 (°C d)	T2 (°C d)	T3 (°C d)	V2 ($\text{cm}^2 \text{cm}^{-2}$) (°C d) ⁻¹
W1N1	1.6284	817.74	414.67	1220.81	1.4277
W1N2	1.3891	900.96	408.52	1393.40	1.2180
W1N3	1.4577	783.78	330.18	1237.39	1.2781
W2N1	1.4446	910.02	435.49	1384.55	1.2667
W2N2	1.3229	891.97	367.64	1416.30	1.1599
W2N3	1.5019	942.29	486.34	1398.25	1.3169
W3N1	1.5235	934.42	485.99	1382.84	1.3358
W3N2	1.5142	835.22	394.14	1276.30	1.3276
W3N3	1.5731	875.63	450.88	1300.39	1.3793

temperature necessary for the N1 treatment decreased in conjunction with an uptick in nitrogen application. T2 was related to the period of potato tuber formation; as the amount of nitrogen applied increased, the effective cumulative temperature decreased. The T3 cumulative temperature was related to the period of potato starch accumulation. Plant growth under the N1 treatment required lesser cumulative temperature; the average growth rate in the rapid period under the N1 treatment was the highest. The maximum growth rate (V1) with an increase in N application was associated with the N3 treatment, and the effective temperature required for maximum growth rate (T1), corresponding to the N1 treatment, was 934.42 °C d. The maximum growth rate (V1) with an increase in N application corresponded to N3, and the effective temperature required for maximum growth rate (T1), corresponding to N1, was 934.42 °C d. The maximum growth rate (V1) with increasing N application corresponded to N3, and the effective temperature required for the maximum growth rate (T1), corresponding to N1, was 934.42 °C d. The T2 cumulative temperature corresponded to the potato tuber formation period, and the growth rate under the N1 treatment was the highest. The T3 cumulative temperature corresponded to the potato starch accumulation period, and starch accumulation under the N1 treatment required the largest cumulative temperature. The rapid period average growth rate was the highest under the N3 treatment. Thus, irrigation and nitrogen application with too much or too little water supply increased the demand for potato LAI for effective cumulative temperature and delayed the growth and development process compared to that under moderate water and nitrogen regulation.

Characteristics of Potato Biomass Changes

Effect of Water and Nitrogen Treatment on Biomass of Potato at Each Reproductive Stage

Figure 7 presents the progression of potato biomass at every stage of reproduction. The patterns of biomass fluctuation under varying water and nitrogen treatments were broadly consistent over both years. The growth of potatoes commenced at a sluggish pace, accelerated, and then once again decelerated, thereby presenting a slow-fast-slow dynamic alteration. This trend was aligned with the biological logistic growth process. During the potato seedling phase, under uniform irrigation quota conditions, the modifications in root, stem, and leaf structure were congruent. They exhibited an increase in response to heightened nitrogen application under the W1 regime, and displayed an initial increase followed by a decrease with escalating nitrogen application under the W2 and W3 treatments. For the amount of N applied, potato stem biomass first increased and then decreased with an increase in the irrigation level under the N1 treatment; the stem biomass increased with an increase in the irrigation level under the W2 treatment, and it first decreased and then increased with an increase in the irrigation level under the W3 treatment. During the potato tuber formation period, the biomass changes in stems and roots followed a pattern similar to seedlings under the same irrigation level. Leaf biomass initially increased and then decreased at various irrigation levels, with the most significant changes

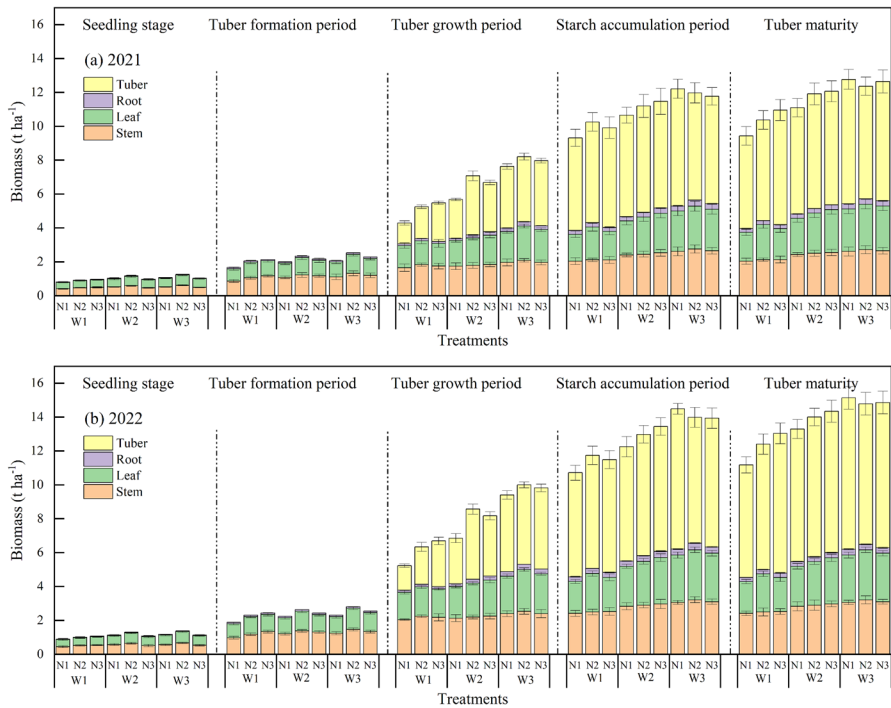


Fig. 7 Biomass change pattern of potato at each reproductive stage

observed under the W3N2 treatment, where the biomass was 51% higher than that under the W1N1 treatment.

In the potato tuber growth period, the changes in root, stem, and leaf biomass were similar when the same amount of nitrogen (N) was applied. They all showed an initial increase and then a decrease with N application under the W1 and W3 treatments, while they increased with the amount of N applied under the W2 treatment. The potato tuber biomass increased with the amount of N applied under the W1 and W3 treatments, and it showed an initial increase and then a decrease with the amount of N applied under the W2 treatment. Regardless of the amount of N applied, all treatments exhibited an increasing trend with an increase in the irrigation quota.

During the potato starch accumulation period, the changes in stem, leaf, and root biomass followed a similar pattern to those recorded in the tuber growth period. The tuber biomass first increased and then decreased with an increase in the amount of N applied under the same irrigation quota for W1, the tuber biomass increased gradually for W2, whereas it first decreased and then increased for W3; the highest tuber biomass was recorded under the W3N2 treatment.

Table 6 presents the analysis of biomass allocation and the significance of various treatments on potatoes at the time of harvest for the years 2021 and 2022. The irrigation rate was highly significant ($P < 0.001$) for all organs of potato biomass, but the interaction of nitrogen application and irrigation rate with the application of

Table 6 The allocation of biomass and significance analysis of different water and nitrogen treatments in the potato harvesting period

Year	Treatment	Stem (t ha ⁻¹)	Leaf (t ha ⁻¹)	Root (t ha ⁻¹)	Tubers (t ha ⁻¹)
2021	W1N1	2.03c	1.70d	0.228d	5.46c
	W1N2	2.12bc	2.07bc	0.235d	5.94bc
	W1N3	2.13bc	1.82cd	0.232d	6.77ab
	W2N1	2.42ab	2.15b	0.249 cd	6.28abc
	W2N2	2.50ab	2.37ab	0.261bcd	6.78ab
	W2N3	2.55a	2.53a	0.286abc	6.70ab
	W3N1	2.61a	2.51a	0.303ab	7.35a
	W3N2	2.71a	2.68a	0.315a	6.65ab
	W3N3	2.65a	2.64a	0.309a	7.04ab
	W	**	**	**	*
	N	ns	*	ns	ns
	W×N	ns	ns	ns	ns
2022	W1N1	2.41d	1.87e	0.240f	6.65c
	W1N2	2.49cd	2.25cde	0.246ef	7.41bc
	W1N3	2.52cd	2.00de	0.261def	8.24ab
	W2N1	2.83bc	2.36bcd	0.278cd	7.83ab
	W2N2	2.89ab	2.57abc	0.274cde	8.25ab
	W2N3	2.96ab	2.73ab	0.300bc	8.35ab
	W3N1	3.06ab	2.79ab	0.345a	8.94a
	W3N2	3.21a	2.95a	0.330a	8.29ab
	W3N3	3.10ab	2.86a	0.325ab	8.57a
	W	**	**	**	**
	N	ns	ns	ns	ns
	W×N	ns	ns	ns	ns

* indicates highly significant, ** significant respectively, ns indicates not significant

nitrogen was not significant for all organs of potato biomass, which indicated that the irrigation rate was more sensitive to potato growth. Stem, leaf, and root biomass was maximum under the W3N2 treatment, and tuber biomass was maximum under the W1N1 treatment at the same irrigation rate.

Dynamics of Total Potato Biomass with Cumulative Temperature and Its Modeling

The nonlinear regression analysis of the logistic model of total potato biomass and the cumulative temperature for each treatment in 2021 was performed using the SPSS 26 software, and the results are presented in Table 7. Our findings showed that the model fit was satisfactory, and the logistic model based on the effective cumulative temperature accurately simulated the dynamic growth of potato LAI under each combination of water and nitrogen treatment.

Table 7 Parameters of the growth dynamic equation for total potato biomass under different water and nitrogen treatments in 2021

Treatment	Parameter			R^2	RMSE (t ha ⁻¹)	NRMSE (%)	d
	K	a	b				
W1N1	1.1844	74.2125	6.3803	0.9710	0.5937	11.6468	0.9936
W1N2	1.1732	49.5127	6.0155	0.9800	0.9504	16.4825	0.9849
W1N3	1.2032	40.3732	5.4418	0.9930	0.3235	5.4923	0.9984
W2N1	1.1618	53.9960	6.1042	0.9870	0.4852	7.9699	0.9967
W2N2	1.1232	37.7622	5.9379	0.9960	0.2918	4.3323	0.9989
W2N3	1.1276	53.6038	6.3129	0.9930	0.3661	5.4882	0.9984
W3N1	1.0826	71.8987	7.1127	0.9960	0.3357	4.6987	0.9988
W3N2	1.0759	46.2644	6.7730	0.9960	0.3084	4.2461	0.9989
W3N3	1.0689	53.4278	6.7996	0.9990	0.1957	2.7419	0.9996

Testing the Model for the Growth of Total Biomass of Potato

The accuracy and precision of the simulated models were tested using the data of total potato biomass and cumulative temperature obtained in 2022. The measured and simulated R^2 values of total potato biomass were greater than 0.9883 (Table 8), which indicated that the model was credible. With the RMSE ranging between 0.2469 and 0.4928 t ha⁻¹, and the NRMSE varying from 3.4813 to 8.1545%, both remaining below 10%, these findings indicate a higher precision in the simulation outcomes. The goodness-of-fit index (d) exceeded 0.9968, which suggests a high degree of alignment between the simulated and observed results, implying superior model performance. Our findings showed that the values and changes were similar under each treatment condition in both years, and the regression equation in any one year might be used for analyzing the data of another year. Additionally, the evaluation coefficients of the W1N3 treatment

Table 8 Evaluation of the measured and simulated values of total potato biomass in 2022

Treatment	RMSE (t ha ⁻¹)	NRMSE (%)	d	R^2
W1N1	0.4928	8.1545	0.9968	0.9883
W1N2	0.4776	6.8564	0.9974	0.9921
W1N3	0.2469	3.4813	0.9993	0.9985
W2N1	0.4127	5.6289	0.9983	0.9950
W2N2	0.3750	4.6757	0.9987	0.9954
W2N3	0.3437	4.2950	0.9990	0.9965
W3N1	0.4303	5.0321	0.9987	0.9947
W3N2	0.4736	5.4044	0.9982	0.9937
W3N3	0.4320	5.0864	0.9985	0.9945

Fig. 8 A 1:1 plot of the measured and simulated total potato biomass in 2022

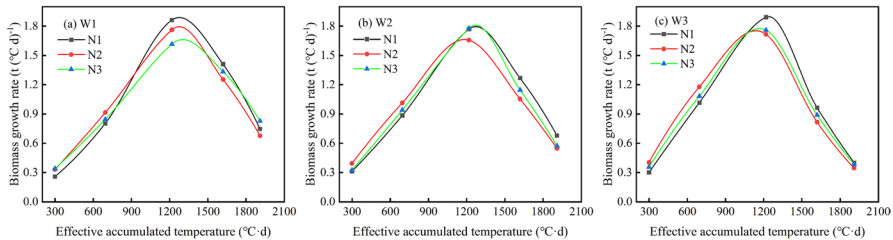
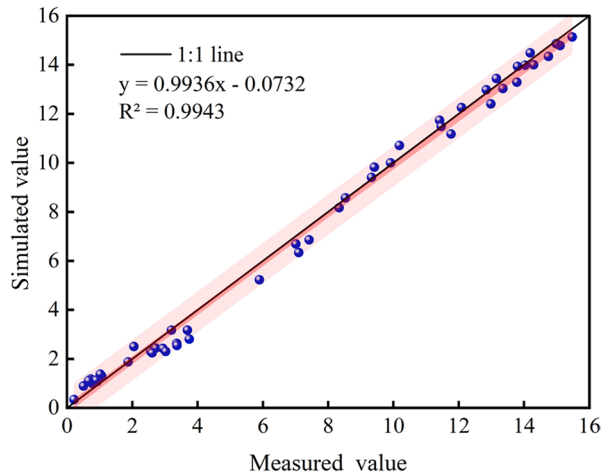


Fig. 9 Change pattern of potato biomass growth rate (2021)

were relatively better under different N fertilizer levels and different irrigation levels, which implied that the model fit was better under irrigation treatment.

Figure 8 presents a 1:1 plot of the measured and simulated values for each treatment equation. As depicted in the figure, the R^2 between the measured and simulated values for the total biomass of the potato was found to be 0.9943. The results showed that the logistic equation obtained from the simulation under this test condition had biological significance and might be used to further analyze the dynamic changes in total potato biomass with the cumulative temperature under different treatments.

Analysis of the Growth Rate of Total Potato Biomass and the Characteristic Parameters of Its Equation

The derivatives of the logistic model, obtained for each regime in 2021, were employed to derive the equation characterizing the growth rate of total potato biomass. This relationship is visually depicted in Fig. 9. At the W1 level, the effective cumulative temperature ranged from 299.30 to 694.60 °C d with $N2 > N3 > N1$, from 694.60 to 1217.35 °C d with $N1 > N2 > N3$, and from 1618.85 to 1908.85 °C d

with N2. Thus, the total biomass growth rate during potato growth and development reached a peak in the range of $N2 > N3 > N1$ at the W2 and W3 levels. The growth rate was closely associated with water and nitrogen treatment.

Table 9 presents the characteristic parameters of the logistic model pertaining to the dynamics of potato biomass in 2021. Under the W1 regime, the peak growth rate (V1) exhibited a declining trend with increasing nitrogen application for each treatment. The time point T1, necessary to reach V1, was highest for the N3 treatment, registering at 1297.22 °C d. The T2 cumulative temperature (ranging from 820.36 to 894.53 °C d) corresponded with the potato block growth phase. The T3 effective cumulative temperature initially decreased and subsequently increased in response to an escalation in nitrogen application. For the potato starch accumulation period, the N2 treatment required a lower cumulative temperature; V2 is the maximum for N1 processing. Under the W2 treatment, the maximum growth rate (V1) of each treatment with increasing nitrogen application was the highest for the N3 treatment. The time point T1, which corresponds to the peak growth rate, was found to be the most substantial for the N1 treatment, clocking in at 1247.38 °C d. The T2 cumulative temperature corresponded to the potato tuber growth period, and the N1 treatment showed the highest value. The T3 cumulative temperature represented the potato starch accumulation period, and the N1 treatment required the highest cumulative temperature. Among all nitrogen treatments under the W3 treatment, the N1 treatment showed the highest maximum growth rate (V1), and the effective cumulative temperature required for T1, corresponding to the N1 treatment, was 1147.36 °C d, the highest among all treatments.

The T2 cumulative temperature corresponded to the potato tuber growth period, and tuber growth was the highest under the N1 treatment. The T3 cumulative temperature was associated with the potato starch accumulation period, and the N1 treatment required the largest cumulative temperature. Overall, the maximum growth rate (V1) was the highest under the W3N1 treatment. The highest T2 cumulative temperature was 1297.22 °C d under the W1N3 treatment. The highest T2 cumulative temperature under the W1N1 treatment was 894.53 °C d. The highest

Table 9 The characteristic parameters of the logistic model for the changes in potato biomass (2021)

Treatment	Rate peak parameter		Rapid increase period parameters		
	V1 t (°C d) ⁻¹	T1 (°C d)	T2 (°C d)	T3 (°C d)	V2 t (°C d) ⁻¹
W1N1	1.8891	1288.54	894.53	1682.54	1.6564
W1N2	1.7644	1238.26	820.36	1656.16	1.5470
W1N3	1.6369	1297.22	835.27	1759.18	1.4353
W2N1	1.7729	1247.38	835.55	1659.21	1.5545
W2N2	1.6674	1167.35	743.99	1590.72	1.4620
W2N3	1.7796	1203.93	805.72	1602.14	1.5604
W3N1	1.9251	1147.36	793.92	1500.79	1.6879
W3N2	1.8218	1080.66	709.49	1451.82	1.5973
W3N3	1.8171	1116.84	747.13	1486.55	1.5932

T3 cumulative temperature under the W1N3 treatment was 1759.18 °C d. The rapid period average growth rate (V2) was the highest under the W3N1 treatment. To summarize, too much or too little irrigation and N application increased the potato biomass demand for the effective cumulative temperature and delayed the growth and development of the potato plant, compared to that under moderate water and N regulation.

Potato Yield and Quality Analysis

Table 10 displays the yield, commercial potato output, and quality of potato tubers for each respective treatment. The rate of irrigation and nitrogen application exerted a significant influence ($P < 0.05$) on the yield, commercial potato output, and tuber quality across both years, with the patterns largely aligning. With nitrogen

Table 10 Yield, commercial potatoes, and quality of potato tubers

Year	Treatments	Tuber yield (t ha ⁻¹)	Commodity tubers (t ha ⁻¹)	Vitamin C content (g 100 g ⁻¹)	Reduc- ing sugar content (g 100 g ⁻¹)	Starch content (g 100 g ⁻¹)
2021	W1N1	39.97g	34.83d	19.76abc	0.19cd	16.30c
	W1N2	43.84f	36.83cd	17.54c	0.27a	20.60a
	W1N3	49.42cd	41.05b	19.09bc	0.19b	16.70bc
	W2N1	46.13ef	39.05bc	19.47abc	0.23cd	20.80a
	W2N2	48.01cde	40.04b	18.12bc	0.13f	19.7abc
	W2N3	50.19bc	41.33b	20.25abc	0.16e	19.20abc
	W3N1	54.53a	47.79a	22.28a	0.20c	20.20ab
	W3N2	46.87de	38.80bc	20.54ab	0.24b	21.20a
	W3N3	52.50ab	46.57a	18.31bc	0.17de	22.40a
	W	**	**	ns	**	**
	N	**	**	ns	**	ns
	W × N	**	**	ns	**	*
2022	W1N1	39.89d	37.66d	22.01c	0.19cd	16.02ab
	W1N2	44.44c	38.07cd	20.34d	0.26a	18.20a
	W1N3	49.46ab	45.36ab	22.61c	0.17cd	15.32bc
	W2N1	47.07bc	40.69bcd	24.29b	0.22b	17.47a
	W2N2	49.50ab	41.08bcd	21.91c	0.18cd	16.3ab
	W2N3	50.11ab	44.79ab	24.78b	0.17d	15.36bc
	W3N1	53.64a	48.53a	26.25a	0.18cd	13.80c
	W3N2	49.75ab	43.58abc	25.59ab	0.19c	14.39bc
	W3N3	51.42ab	46.08ab	24.85b	0.17cd	16.35a
	W	**	**	**	**	**
	N	*	**	**	**	*
	W × N	**	**	**	**	**

* indicates highly significant, ** significant respectively, ns indicates not significant

application held constant, both low and high nitrogen treatments demonstrated a yield increase corresponding with escalating irrigation, while the medium nitrogen treatment exhibited an initial increase followed by a decrease. The highest total yield was recorded for the W3N1 treatment in both years, registering an average yield of 54.09 tons per hectare and an average commercial potato yield of 48.16 tons per hectare. This indicated that greater irrigation levels and lower N application presented the highest yield, while excessive N application significantly decreased the yield. The changes in potato quality in both years were similar.

The pattern of changes in potato quality over the 2 years was largely consistent. When applying an identical amount of water, the vitamin C (VC) content in potatoes under the W1 and W2 regimes showed a tendency to decrease initially and then increase with the escalation of nitrogen application. Conversely, the W3 regime displayed an initial increase followed by a decrease. When the same quantity of nitrogen was applied, the N2 regime showed a steady increase, whereas the N3 regime exhibited a decrease followed by an increase. The maximum VC content was observed in the W3N1 treatment across different regimes. The alterations in the reducing sugar content of potatoes under different water and nitrogen conditions were significant, with the highest content observed in the W1N2 regime. The starch content in potatoes from each treatment reached its peak under high water and high nitrogen conditions, recording an average increase of 23.30% compared to the lowest water and nitrogen application levels. In summary, both water and nitrogen treatments significantly impacted the quality of potatoes. Total yield, commercial potato yield, reducing sugar, and starch increased by 4.09%, 3.97%, 10.68%, and 12.43%, respectively, and content decreased by 12.51% in N1 compared to N3 under conventionally irrigated W3 treatment. As potato caused water stress, total and commercial potato yields gradually increased with increasing nitrogen application; potato was more affected by irrigation than by nitrogen application when regular irrigation was applied. Excessive nitrogen application decreased potato vitamin C and reducing sugar and increased potato starch content. This study concluded that W3N1 was the best treatment.

Correlation Between Potato Yield and Growth Indicators

The correlation between the potato yield and LAI during starch accumulation under different treatments is shown in Fig. 10a and b. The biennial data for LAI and yield during the starch accumulation phase in potato adhered to the one-dimensional linear regression equation, respectively: $y = 5.2369x + 20.757$ and $y = 3.8173x + 25.589$, respectively, with R^2 values of 0.6994 and 0.7056 for the linear fit. The correlation between the yield and biomass of potatoes under different treatments is shown in Fig. 10c and d. The potato biomass data accumulated over the 2-year period conformed to the one-dimensional linear regression equation: $y = 3.4431x + 8.315$ and $y = 2.9401x + 8.184$, respectively, with R^2 values of 0.7710 and 0.8877 for the linear fit. The 2-year fit showed that the biomass was greater than the LAI, which indicated that the potato yield can be estimated from LAI and biomass, and reasonable control of these two indicators within a certain range can improve the yield.

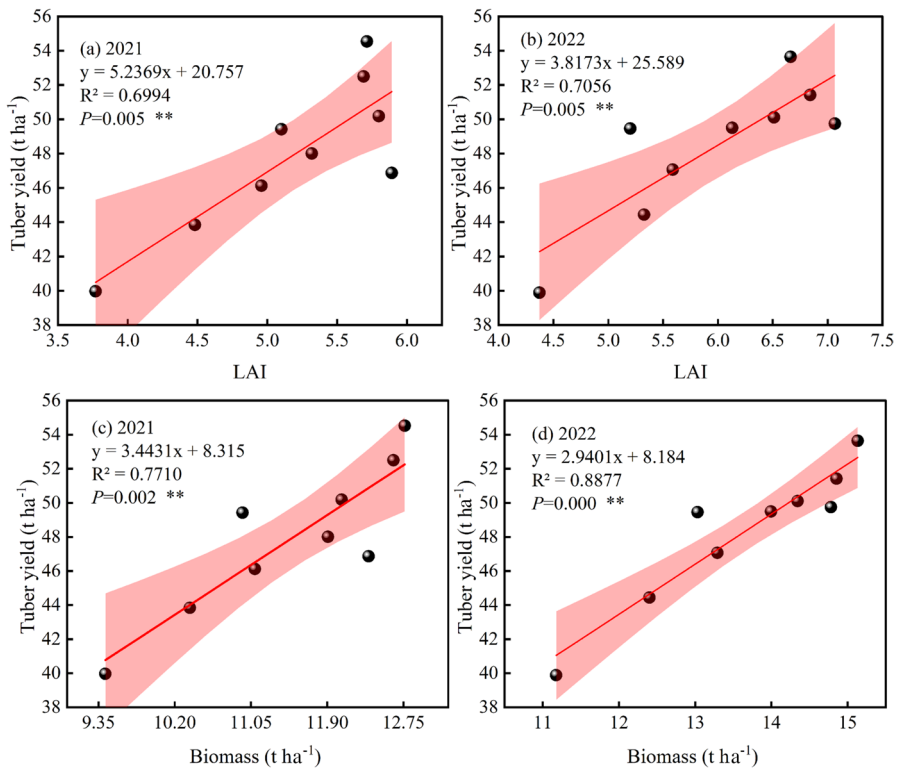


Fig. 10 The correlation of potato yield with the LAI and biomass

Discussion

Effect of Water and Nitrogen Regulation on Potato LAI and Biomass

LAI serves as an indicator of yield formation and provides insights into the dynamic traits and vitality of potato growth (Jonckheere et al. 2004). Employing appropriate combinations of water and nitrogen is crucial in enhancing crop yield and optimizing nitrogen utilization efficiency. However, excessive application of water and nitrogen can lead to nutrient loss (Shen et al. 2023). When the irrigation water amount remained constant, the LAI exhibited variations across different treatments with increasing nitrogen application, demonstrating that nitrogen application can enhance LAI (Bangemann et al. 2014). The increase in the LAI with the intensity of irrigation under the same conditions of N application indicated that irrigation had a greater effect on potato LAI (Camargo et al. 2015). Excessive N application under drip irrigation can decrease the LAI, which is detrimental to potato growth; increasing the irrigation rate appropriately can increase the LAI (Fan et al. 2020).

Potato biomass serves as a significant indicator of growth and can indirectly reflect the potential yield magnitude (Liu et al. 2010). It also guides crop

production management. Accurate information on crop biomass is required for crop yield estimation, crop growth monitoring, crop field management, and control (Zheng et al. 2004). Potatoes can alter biomass allocation when grown under water or N stress, which decreases the yield (Belanger et al. 2001). At the W1 irrigation quota level, potato biomass exhibited an increase alongside a rise in the applied nitrogen (N) amount, which aligns with the findings of a previous study by Lee et al. (2017). This study also noted an increase in biomass with N application and highlighted that an appropriate elevation in N content can enhance crop growth. However, under the W2 and W3 irrigation quotas, the trend of initial biomass increase followed by a decrease with escalating N application concurred with the observations of Zhai et al. (2022). Their research highlighted that dry matter accumulation rose initially and then plateaued as the N quantity increased, indicating that excessive N application hampers crop growth, and high fertilizer levels can inhibit potato growth (Yang et al. 2017). A reasonable irrigation and fertilizer regime can promote healthy crop growth (Li et al. 2021a, b, c).

Model Analysis of Potato LAI and Biomass Based on the Effective Cumulative Temperature

Effective cumulative temperature (GDD), initially introduced in the seventeenth century, serves as an ecosystem metric for examining the daily temperature fluctuations and assessing their impact on various stages of plant growth. This concept provides a foundation for developing climate-adaptive crops in the future (Anandhi 2016). The effective temperature accumulation corresponds to the heat accumulated by the crop during a specific reproductive stage, serving as an indicator of the thermal requirements necessary for crop development during that reproductive period. It is more stable than the number of days in the reproductive period. Therefore, the effective accumulated temperature can be used instead of the number of days after sowing to model crop growth and predict crop yield more accurately (Justin et al. 2013). With the increase in climate change, the influence of the effective cumulative temperature on crop growth is second only to irrigation (Wu et al. 2022). The potato LAI exhibited a strong correlation with the effective cumulative temperature, with LAI increasing as the effective cumulative temperature rose. The maximum LAI value for potatoes was achieved at 1700 °C d. The logistic model was validated using the LAI data of 2 years and found to be a good fit, which agreed with the results of some studies (Yu et al. 2002), suggesting that the logistic model can be used to simulate potato LAI to predict potato growth. Logistic models can simulate better results for LAI and require only a few input variables under near-optimal conditions (Setiyono et al. 2008). The agreement between the simulated and measured values was excellent, as indicated by the evaluation of the logistic growth rate curve using the root mean square error (RMSE). The logistic model demonstrated greater accuracy in simulating both plant LAI and biomass (Xiangxiang et al. 2014). Lower effective cumulative temperatures can result in lower crop yields (Zhang et al. 2022a, b). Soil moisture and effective temperature during tuber formation are the key factors influencing the total potato yield, with soil moisture exerting a more

significant impact than effective temperature. Hence, in cases where the effective cumulative temperature is inadequate, increasing the irrigation quota can be a viable approach to achieve higher potato yield and income while efficiently utilizing water and heat resources (Gu et al. 2022).

Analysis of Potato Yield, Quality, and Related Relationships among Different Treatments

Optimization of irrigation and fertilization management is necessary for improving the yield of maize and resolving the problem of water scarcity (Li et al. 2021a, b, c; Selim et al. 2009). Crop development and yield formation are dynamic and complex processes (Li et al. 2021a, b, c). According to the study conducted by Yan et al. (2022), the yield and irrigation water use efficiency exhibited an increasing trend followed by stabilization with the escalation of N application at a constant irrigation rate. The productivity of the main root yield and sugar yield of sugar beet decreased with an increase in N application, and unreasonable N application also resulted in lower yields (Wasaya et al. 2017). The results revealed that increasing nitrogen application led to an enhancement in yield. Similarly, increasing the irrigation amount contributed to improved yield. However, excessive irrigation had an adverse impact, leading to a reduction in yield. Our results agreed with those of She et al. (2023) who showed that higher N application adversely affected the growth and yield of crop grains. The yield decreased significantly with an increase in N application, and it first increased then decreased with an increase in irrigation (Wang et al. 2022). The investigation conducted by Li et al. (2023a, b) revealed that both total potato yield and commercial yield exhibited an initial increase followed by a decrease with the augmentation of N application. The findings of this study indicated that potato quality was significantly influenced by irrigation rates and N application. This suggests that appropriate levels of irrigation and N application can enhance yield while preserving the crop's quality (Hou et al. 2021). The W3N1 treatment showed the highest VC content in potatoes, which indicated that a high water content and a low level of N significantly increased quality (Stamatiadis et al. 2016). The significant changes in the content of reducing sugars and starch in potatoes under water and nitrogen treatment further indicated that appropriate irrigation can improve the quality of crops (Ballester et al. 2021; Wang et al. 2020, 2021). We also found a significant positive correlation between potato LAI and biomass and the yield through the linear fitting of the data, which indicated that potato yield can be estimated from potato LAI and biomass, and reasonable control of these three indicators within a certain range can effectively improve the yield (Sandhu et al. 2013; Zhang et al. 2004). In the results of this experimental study under higher irrigation volume, it helps N fertilizer transport, thus improving nitrogen use efficiency, ensuring potato growth, and increasing potato tuber yield (Badr et al. 2023). Growing potatoes in arid climates can result in greater yields while conserving irrigation water and reducing nitrogen supply, but the interaction between these two factors must be exploited to maximize resource use efficiency (Badr et al. 2012).

Through the 2-year experimental analysis of the results which are basically consistent, water and nitrogen regulation on potato LAI, biomass, and yield of significant impact indicates that the results of the study can be reliable, and further validates the applicability of the logistic model and the underlying parameters, which concluded that the water and nitrogen regulation on the dry zone of Ningxia under the membrane drip irrigation of potato growth dynamics and the formation of yield process is closely related to the quantitative simulation of the effective cumulative temperature potato. The parameters were established for the quantitative simulation of potato growth and development dynamics by effective cumulative temperature. It provides a scientific theoretical basis for irrigation quota and nitrogen application, and promotes the steady improvement of potato yield.

Conclusion

LAI and biomass exhibited a ‘slow-fast-slow’ dynamic change process with effective accumulated temperature under different water and nitrogen regulations. The LAI of potato reached its maximum at 1700 °C d, with the highest values observed at an irrigation rate of 2100 m³ ha⁻¹ and an N application rate of 190 kg ha⁻¹. The effective temperature (T1) required to reach the maximum growth rate was greatest in the 190 kg ha⁻¹ treatment, at 900.96 °C d, while the maximum growth rate (V1) was greatest in the 270 kg ha⁻¹ treatment, with the effective temperature (T1) being 942.29 °C d. The potato growth rate peaked at 900 °C d, and the effective temperature required for the maximum growth rate (T1) was largest in the 270 kg ha⁻¹ treatment, at 942.29 °C d. The temperature required for the total biomass growth rate to reach its peak during the growth and development period of potato varied among treatments, with the growth rate being closely related to water and nitrogen conditions. Under different water and nitrogen conditions, the average total potato yield and commercial potato yield were highest in the treatments with an irrigation rate of 2100 m³ ha⁻¹ and a nitrogen application rate of 110 kg ha⁻¹, with average yields of 54.09 t ha⁻¹ and 48.16 t ha⁻¹, respectively, and a commercial potato rate of 89.06%. The highest VC content was recorded in potatoes treated with an irrigation rate of 2100 m³ ha⁻¹ and an N application rate of 110 kg ha⁻¹. Reducing sugar content was highest in the treatment with an irrigation rate of 1260 m³ ha⁻¹ and an N application rate of 190 kg ha⁻¹. Potato starch content was highest in the treatment with an irrigation rate of 2100 m³ ha⁻¹ and an N application rate of 270 kg ha⁻¹. The linear relationship between LAI and biomass of potato and yield can be analyzed indirectly, and the logistic model can be used to quantitatively analyze the growth dynamics of drip-irrigated potato in the dry zone of Ningxia using the effective accumulated temperature. This provides a theoretical basis for the dynamic simulation and accurate prediction of crop growth.

Author Contribution Yingpan Yang: data curation, writing—original draft preparation. Na Zhang: conceptualization, methodology. Sheng Li and Wei Tian: software. Yinhao Zhu: supervision.

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Data Availability Data will be made available on request.

Declarations

Competing Interests The authors declare no competing interests.

References

- Anandhi A (2016) Growing degree days – ecosystem indicator for changing diurnal temperatures and their impact on corn growth stages in kansas. *Ecol Indic* 61:149–158. <https://doi.org/10.1016/j.ecoli.2015.08.023>
- Ayadi M, Chiab N, Charfeddine S, Abdelhedi R, Dabous A, Talbi O, Mieulet D, Guiderdoni E, Aifa S, Gargouri-Bouazid R, Hanana M (2022) Improved growth and tuber quality of transgenic potato plants overexpressing either nhx antiporter, clc chloride channel, or both. *Plant Physiol Bioch* 189:46–58. <https://doi.org/10.1016/j.plaphy.2022.07.030>
- Badr MA, El-Tohamy WA, Zaghloul AM (2012) Yield and water use efficiency of potato grown under different irrigation and nitrogen levels in an arid region. *Agr Water Manage* 110:9–15. <https://doi.org/10.1016/j.agwat.2012.03.008>
- Badr MA, Ali E, Salman SR (2023) Effect of nitrogen application and fertigation scheduling on potato yield performance under drip irrigation system. *Gesunde Pflanz*. <https://doi.org/10.1007/s10343-023-00871-y>
- Ballester C, Hornbuckle J, Brinkhoff J, Quayle WC (2021) Effects of three frequencies of irrigation and nitrogen rates on lint yield, nitrogen use efficiency and fibre quality of cotton under furrow irrigation. *Agr Water Manage* 248:106783. <https://doi.org/10.1016/j.agwat.2021.106783>
- Bangemann L, Sieling K, Kage H (2014) The effect of nitrogen and late blight on crop growth, solar radiation interception and yield of two potato cultivars. *Field Crops Res* 155:56–66. <https://doi.org/10.1016/j.fcr.2013.09.022>
- Belanger G, Walsh JR, Richards JE, Milburn PH, Ziadi N (2001) Tuber growth and biomass partitioning of two potato cultivars grown under different n fertilization rates with and without irrigation. *Am J Potato Res* 78(2):109–117. <https://doi.org/10.1007/BF02874766>
- Camargo DC, Montoya F, Córcoles JJ, Ortega JF (2015) Modeling the impacts of irrigation treatments on potato growth and development. *Agr Water Manage* 150:119–128. <https://doi.org/10.1016/j.agwat.2014.11.017>
- Chen WQ, Wu LR, Liu TG, Xu SC, Jin SL, Peng YL, Wang BT (2009) Race dynamics, diversity, and virulence evolution in puccinia striiformis f. Sp. Tritici, the causal agent of wheat stripe rust in china from 2003 to 2007. *Plant Dis* 93(11):1093–1101. <https://doi.org/10.1094/PDIS-93-11-1093>
- Chu L, Huang C, Liu Q, Liu G (2016) Estimation of winter wheat phenology under the influence of cumulative temperature and soil salinity in the yellow river delta, china, using modis time-series data. *Int J Remote Sens* 37(10):2211–2232. <https://doi.org/10.1080/01431161.2015.1131871>
- Eeswaran R, Nejadhashemi AP, Kpodo J, Curtis ZK, Adhikari U, Liao H, Li S, Hernandez-Suarez JS, Alves FC, Raschke A, Jha PK (2021) Quantification of resilience metrics as affected by conservation agriculture at a watershed scale. *Agr Ecosyst Environ* 320:107612. <https://doi.org/10.1016/j.agee.2021.107612>
- Fan J, Lu X, Gu S, Guo X (2020) Improving nutrient and water use efficiencies using water-drip irrigation and fertilization technology in northeast china. *Agr Water Manage* 241:106352. <https://doi.org/10.1016/j.agwat.2020.106352>
- Fiers M, Edel-Hermann V, Chatot C, Le Hingrat Y, Alabouvette C, Steinberg C (2012) Potato soil-borne diseases. A review. *Agron Sustain Dev* 32(1):93–132. <https://doi.org/10.1007/s13593-011-0035-z>
- Giacomini Sari B, Dal Col Lúcio A, Souza Santana C, Olivoto T, Diel MI, Ketzer Krysczun D (2019) Nonlinear growth models: an alternative to anova in tomato trials evaluation. *Eur J Agron* 104:21–36. <https://doi.org/10.1016/j.eja.2018.12.012>

- Gu H, Fangfang M, Nan W, Youming M, Fugui Z, Long Z, Xianqing H (2022) Effects of tillage with mulching on potato yield and the characteristics of soil water and temperature in arid area of southern ningxia. *Chin J Appl Ecol* 33(12):3352–3362
- Hou X, Fan J, Hu W, Zhang F, Yan F, Xiao C, Li Y, Cheng H (2021) Optimal irrigation amount and nitrogen rate improved seed cotton yield while maintaining fiber quality of drip-fertigated cotton in northwest china. *Ind Crop Prod* 170:113710. <https://doi.org/10.1016/j.indcrop.2021.113710>
- Hu X, Luo H, Guo M, Wang J (2022) Ecological technology evaluation model and its application based on logistic regression. *Ecol Indic* 136:108641. <https://doi.org/10.1016/j.ecolind.2022.108641>
- Ierna A, Mauromicale G (2018) Potato growth, yield and water productivity response to different irrigation and fertilization regimes. *Agr Water Manage* 201:21–26. <https://doi.org/10.1016/j.agwat.2018.01.008>
- Jahangirlou MR, Morel J, Akbari GA, Alahdadi I, Soufizadeh S, Parsons D (2023) Combined use of apsim and logistic regression models to predict the quality characteristics of maize grain. *Eur J Agron* 142:126629. <https://doi.org/10.1016/j.eja.2022.126629>
- Jonckheere I, Fleck S, Nackaerts K, Muys B, Coppin P, Weiss M, Baret F (2004) Review of methods for in situ leaf area index determination. *Agr Forest Meteorol* 121(1–2):19–35. <https://doi.org/10.1016/j.agrformet.2003.08.027>
- Justin VW, Kersebaum KC, Peng S, Milner M, Cassman KG (2013) Estimating crop yield potential at regional to national scales. *Field Crop Res* 143:34–43. <https://doi.org/10.1016/j.fcr.2012.11.018>
- Lee M, Wycislo A, Guo J, Lee DK, Voigt T (2017) Nitrogen fertilization effects on biomass production and yield components of miscanthus $\times$ giganteus. *Front Plant Sci* 8:544. <https://doi.org/10.3389/fpls.2017.00544>
- Li Y, Lu W, Lyu D, Su F, Liu S, Li H, Wang X, Liu Z, Hu L (2018) Effects of different nitrogen application rates on starch accumulation, starch synthase gene expression and enzyme activity in two distinctive potato cultivars. *Potato Res* 61(4):309–326. <https://doi.org/10.1007/s11540-018-9379-y>
- Li R, Guo J, Song Y (2020) Optimizing parameters of a non-linear accumulated temperature model and method to calculate linear accumulated temperature for spring maize in northeast china. *Theor Appl Climatol* 141(3–4):1629–1644. <https://doi.org/10.1007/s00704-020-03279-0>
- Li C, Wang Y, Ma C, Chen W, Li Y, Li J, Ding F, Xiao Z (2021a) Improvement of wheat grain yield prediction model performance based on stacking technique. *Appl Sci* 11(24):12164. <https://doi.org/10.3390/app112412164>
- Li H, Mei X, Wang J, Huang F, Hao W, Li B (2021b) Drip fertigation significantly increased crop yield, water productivity and nitrogen use efficiency with respect to traditional irrigation and fertilization practices: a meta-analysis in china. *Agr Water Manage* 244:106534. <https://doi.org/10.1016/j.agwat.2020.106534>
- Li J, Zhang Z, Yao C, Liu Y, Wang Z, Fang B, Zhang Y (2021c) Improving winter wheat grain yield and water-/nitrogen-use efficiency by optimizing the micro-sprinkling irrigation amount and nitrogen application rate. *J Integr Agr* 20(2):606–621. [https://doi.org/10.1016/S2095-3119\(20\)63407-4](https://doi.org/10.1016/S2095-3119(20)63407-4)
- Li S, Chen H, Jiang S, Hu F, Xing D, Du B (2023a) Selenium and nitrogen fertilizer management improves potato root function, photosynthesis, yield and selenium enrichment. *Sustainability-Basel* 15(7):6060. <https://doi.org/10.3390/su15076060>
- Li Z, Huang H, Duan Z, Zhang W (2023b) Control temperature of greenhouse for higher yield and higher quality grapes production by combining stb in situ service with on time sensor monitoring. *Heliyon* 9(2):e13521. <https://doi.org/10.1016/j.heliyon.2023.e13521>
- Liao Z, Zheng J, Fan J, Pei S, Dai Y, Zhang F, Li Z (2022) Novel models for simulating maize growth based on thermal time and photothermal units: applications under various mulching practices. *J Integr Agr*. <https://doi.org/10.1016/j.jia.2022.08.018>
- Liu J, Pattey E, Miller JR, Mcnairn H, Smith A, Hu B (2010) Estimating crop stresses, aboveground dry biomass and yield of corn using multi-temporal optical data combined with a radiation use efficiency model. *Remote Sens Environ* 114(6):1167–1177. <https://doi.org/10.1016/j.rse.2010.01.004>
- Lo TH, Rudnick DR, Burr CA, Stockton MC, Werle R (2019) Approaches to evaluating grower irrigation and fertilizer nitrogen amount and timing. *Agr Water Manage* 213:693–706. <https://doi.org/10.1016/j.agwat.2018.11.010>
- Ma T, Zeng W, Lei G, Wu J, Huang J (2021) Predicting the rooting depth, dynamic root distribution and the yield of sunflower under different soil salinity and nitrogen applications. *Ind Crop Prod* 170:113749. <https://doi.org/10.1016/j.indcrop.2021.113749>
- Meise P, Seddig S, Uptmoor R, Ordon F, Schum A (2019) Assessment of yield and yield components of starch potato cultivars (*solanum tuberosum* l.) Under nitrogen deficiency and drought stress conditions. *Potato Res* 62(2):193–220. <https://doi.org/10.1007/s11540-018-9407-y>

- Meno L, Escuredo O, Rodríguez-Flores MS, Seijo MC (2020) Modification of the tomcast model with aerobiological data for management of potato early blight. *Agronomy* 10(12):1872. <https://doi.org/10.3390/agronomy10121872>
- Mohanty M, Painuli DK, Misra AK, Bandyopadhyaya KK, Ghosh PK (2006) Estimating impact of puddling, tillage and residue management on wheat (*triticum aestivum*, L.) Seedling emergence and growth in a rice–wheat system using nonlinear regression models. *Soil Tillage Res* 87(1):119–130. <https://doi.org/10.1016/j.still.2005.03.002>
- Qi D, Hu T, Liu T (2020) Biomass accumulation and distribution, yield formation and water use efficiency responses of maize (*zea mays* L.) To nitrogen supply methods under partial root-zone irrigation. *Agr Water Manage* 230:105981. <https://doi.org/10.1016/j.agwat.2019.105981>
- Rady MOA, Semida WM, Howladar SM, Abd El-Mageed TA (2021) Raised beds modulate physiological responses, yield and water use efficiency of wheat (*triticum aestivum* L) under deficit irrigation. *Agr Water Manage* 245:106629. <https://doi.org/10.1016/j.agwat.2020.106629>
- Sandhu N, Jain S, Kumar A, Mehla BS, Jain R (2013) Genetic variation, linkage mapping of qtl and correlation studies for yield, root, and agronomic traits for aerobic adaptation. *Bmc Genet* 14:104. <https://doi.org/10.1186/1471-2156-14-104>
- Scott GJ, Petsakos A, Juarez H (2019) Climate change, food security, and future scenarios for potato production in india to 2030. *Food Secur* 11(1):43–56. <https://doi.org/10.1007/s12571-019-00897-z>
- Selim EM, Mosa AA, El-Ghamry AM (2009) Evaluation of humic substances fertigation through surface and subsurface drip irrigation systems on potato grown under Egyptian sandy soil conditions. *Agr Water Manage* 96(8):1218–1222. <https://doi.org/10.1016/j.agwat.2009.03.018>
- Sepaskhah AR, Fahandezh-Saadi S, Zand-Parsa S (2011) Logistic model application for prediction of maize yield under water and nitrogen management. *Agr Water Manage* 99(1):51–57. <https://doi.org/10.1016/j.agwat.2011.07.019>
- Setiyono TD, Weiss A, Specht JE, Cassman KG, Dobermann A (2008) Leaf area index simulation in soybean grown under near-optimal conditions. *Field Crop Res* 108(1):82–92. <https://doi.org/10.1016/j.fcr.2008.03.005>
- Shabani A, Sepaskhah AR, Kamgar-Haghighi AA (2014) Estimation of yield and dry matter of rapeseed using logistic model under water salinity and deficit irrigation. *Archiv Für Acker- und Pflanzenbau und Bodenkunde* 60(7):951–969. <https://doi.org/10.1080/03650340.2013.858807>
- She Y, Li P, Qi X, Rahman SU, Guo W (2023) Effects of nitrogen application on winter wheat growth, water use, and yield under different shallow groundwater depths. *Front Plant Sci* 14:1114611. <https://doi.org/10.3389/fpls.2023.1114611>
- Shen H, Gao Y, Sun K, Gu Y, Ma X (2023) Effects of differential irrigation and nitrogen reduction replacement on winter wheat yield and water productivity and nitrogen-use efficiency. *Agr Water Manage* 282:108289. <https://doi.org/10.1016/j.agwat.2023.108289>
- Silva ALBR, Zotarelli L, Dukes MD, van Santen E, Asseng S (2023) Nitrogen fertilizer rate and timing of application for potato under different irrigation methods. *Agr Water Manage* 283:108312. <https://doi.org/10.1016/j.agwat.2023.108312>
- Stamatiadis S, Tsadiras C, Samaras V, Schepers JS, Eskridge K (2016) Nitrogen uptake and n-use efficiency of mediterranean cotton under varied deficit irrigation and n fertilization. *Eur J Agron* 73:144–151. <https://doi.org/10.1016/j.eja.2015.11.013>
- Tang Z, Guo J, Xiang Y, Lu X, Wang Q, Wang H, Cheng M, Wang H, Wang X, An J, Abdelghany A, Li Z, Zhang F (2022) Estimation of leaf area index and above-ground biomass of winter wheat based on optimal spectral index. *Agronomy* 12(7):1729. <https://doi.org/10.3390/agronomy12071729>
- Wang H, Wang X, Bi L, Wang Y, Fan J, Zhang F, Hou X, Cheng M, Hu W, Wu L, Xiang Y (2019) Multi-objective optimization of water and fertilizer management for potato production in sandy areas of northern china based on topsis. *Field Crop Res* 240:55–68. <https://doi.org/10.1016/j.fcr.2019.06.005>
- Wang X, Guo T, Wang Y, Xing Y, Wang Y, He X (2020) Exploring the optimization of water and fertilizer management practices for potato production in the sandy loam soils of northwest china based on pca. *Agr Water Manage* 237:106180. <https://doi.org/10.1016/j.agwat.2020.106180>
- Wang H, Cheng M, Zhang S, Fan J, Feng H, Zhang F, Wang X, Sun L, Xiang Y (2021) Optimization of irrigation amount and fertilization rate of drip-fertigated potato based on analytic hierarchy process and fuzzy comprehensive evaluation methods. *Agr Water Manage* 256:107130. <https://doi.org/10.1016/j.agwat.2021.107130>
- Wang H, Xiang Y, Zhang F, Tang Z, Guo J, Zhang X, Hou X, Wang H, Cheng M, Li Z (2022) Responses of yield, quality and water-nitrogen use efficiency of greenhouse sweet pepper to different drip fertigation regimes in northwest china. *Agr Water Manage* 260:107279. <https://doi.org/10.1016/j.agwat.2021.107279>

- Wasaya A, Tahir M, Ali H, Hussain M, Yasir TA, Sher A, Ijaz M, Sattar A (2017) Influence of varying tillage systems and nitrogen application on crop allometry, chlorophyll contents, biomass production and net returns of maize (*zea mays* l.). *Soil Tillage Res* 170:18–26. <https://doi.org/10.1016/j.still.2017.02.006>
- Wu F, Qiu Y, Huang W, Guo S, Han Y, Wang G, Li X, Lei Y, Yang B, Xiong S, Xin M, Chen J, Wang Z, Feng L, Li Y (2022) Water and heat resource utilization of cotton under different cropping patterns and their effects on crop biomass and yield formation. *Agr Forest Meteorol* 323:109091. <https://doi.org/10.1016/j.agrformet.2022.109091>
- Xiangxiang W, Quanjiu W, Jun F, Lijun S, Xinlei S (2014) Logistic model analysis of winter wheat growth on china's loess plateau. *Can J Plant Sci* 94(8):1471–1479. <https://doi.org/10.4141/cjps2013-293>
- Xing Y, Niu X, Wang N, Jiang W, Gao Y, Wang X (2020) The correlation between soil nutrient and potato quality in loess plateau of china based on pls-r. *Sustainability-Basel* 12(4):1588. <https://doi.org/10.3390/su12041588>
- Xing Y, Zhang T, Jiang W, Li P, Shi P, Xu G, Cheng S, Cheng Y, Fan Z, Wang X (2022) Effects of irrigation and fertilization on different potato varieties growth, yield and resources use efficiency in the northwest china. *Agr Water Manage* 261:107351. <https://doi.org/10.1016/j.agwat.2021.107351>
- Yan F, Liu X, Bai W, Fan J, Zhang F, Xiang Y, Hou X, Pei S, Dai Y, Zeng H, Wang Y (2022) Multi-objective optimization of water and nitrogen regimes for drip-fertigated sugar beet in a desert climate. *Field Crop Res* 288:108703. <https://doi.org/10.1016/j.fcr.2022.108703>
- Yang K, Wang F, Shock CC, Kang S, Huo Z, Song N, Ma D (2017) Potato performance as influenced by the proportion of wetted soil volume and nitrogen under drip irrigation with plastic mulch. *Agr Water Manage* 179:260–270. <https://doi.org/10.1016/j.agwat.2016.04.014>
- Yang Y, Yin J, Ma Z, Wei X, Sun F, Yang Z (2023) Water and nitrogen regulation effects and system optimization for potato (*solanum tuberosum* l.) Under film drip irrigation in the dry zone of ningxia china. *Agronomy* 13(2):308. <https://doi.org/10.3390/agronomy13020308>
- Yin S, Li P, Xu Y, Xue L, Hao D, Liu J, Yang T, Yang Z, Xu C (2018) Logistic model-based genetic analysis for kernel filling in a maize RIL population. *Euphytica* 214:1–16. <https://doi.org/10.1007/s10681-018-2162-y>
- Yu Q, Liu J, Zhang Y, Li J (2002) Simulation of rice biomass accumulation by an extended logistic model including influence of meteorological factors. *Int J Biometeorol* 46(4):185–191. <https://doi.org/10.1007/s00484-002-0141-3>
- Yuchuan F, Xudong Y, Hongfei Z, Sheng PH, Lining L, Tianshu W, Shujuan L, Yangliu Y, Qiang Z, Jianchu S (2017) Suitable evaluation for crops and water-saving irrigation methods for arable sandy land in Inner Mongolia. *Trans Chin Soc Agric Eng* 33(21):115–127
- Zhai J, Zhang G, Zhang Y, Xu W, Xie R, Ming B, Hou P, Wang K, Xue J, Li S (2022) Effect of the rate of nitrogen application on dry matter accumulation and yield formation of densely planted maize. *Sustainability-Basel* 14(22):14940. <https://doi.org/10.3390/su142214940>
- Zhang Z, Li P, Wang L, Hu Z, Zhu L, Zhu Y (2004) Genetic dissection of the relationships of biomass production and partitioning with yield and yield related traits in rice. *Plant Sci* 167(1):1–8. <https://doi.org/10.1016/j.plantsci.2004.01.007>
- Zhang R, Yang D, Li S, Chen J, Hu D, Guo H, Wang C, Wang Y, Cong X (2022a) Coupling of environmental factors and growth stages in simulation of maize biomass allocation. *Plant Soil*. <https://doi.org/10.1007/s11104-022-05794-7>
- Zhang Z, Zhou N, Xing Z, Liu B, Tian J, Wei H, Gao H, Zhang H (2022b) Effects of temperature and radiation on yield of spring wheat at different latitudes. *Agriculture* 12(5):627. <https://doi.org/10.3390/agriculture12050627>
- Zheng D, Rademacher J, Chen J, Crow T, Bresee M, Le Moine J, Ryu S (2004) Estimating aboveground biomass using landsat 7 etm+ data across a managed landscape in northern wisconsin, usa. *Remote Sens Environ* 93(3):402–411. <https://doi.org/10.1016/j.rse.2004.08.008>

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